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I. Multiple Choice Questions

1. One of the combinations from the fundamental physical constants is $\frac{hc}{G}$. The unit of this expression is

a) kg^2 b) m^3

c) s^{-1} d) m

2. If the error in the measurement of radius is 2%, then the error in the determination of volume of the sphere will be

a) 8% b) 2%

c) 4% d) 6%

3. If the length and time period of an oscillating pendulum have errors of 1% and 3% respectively then the error in measurement of acceleration due to gravity is

a) 4% [Related to AMPMT 2008]

b) 5%

c) 6%

d) 7%

4. The length of a body is measured as 3.51 m, if the accuracy is 0.01mm, then the percentage error in the measurement is

a) 351% b) 1%

c) 0.28% d) 0.035%

5. Which of the following has the highest number of significant figures?

a) 0.007 m^2

b) 2.64×10^{24} kg

c) 0.0006032 m^2

d) 6.3200 J

6. If $\pi = 3.14$, then the value of π^2 is

a) 9.8596 b) 9.860

c) 9.86 d) 9.9

7. Which of the following pairs of physical quantities have same dimension?

a) force and power

b) torque and energy

c) torque and power

d) force and torque

8. The dimensional formula of Planck's constant h is [AMU, Main, JEE, NEET]

a) $[ML^2T^{-1}]$ b) $[ML^2T^{-3}]$

c) $[MLT^{-1}]$ d) $[ML^3T^{-3}]$

9. The velocity of a particle v at an instant t is given by $v = at + bt^2$. The dimensions of b is

a) [L] b) $[LT^{-1}]$

c) $[LT^{-2}]$ d) $[LT^{-3}]$

10. The dimensional formula for gravitational constant G is [Related to AIPMT 2004]

a) $[ML^3T^{-2}]$ b) $[M^{-1}L^3T^{-2}]$

c) $[M^{-1}L^{-3}T^{-2}]$ d) $[ML^{-3}T^2]$

11. The density of a material in CGS system of units is 4 g cm^{-3}. In a system of units in which unit of length is 10 cm and unit of mass is 100 g, then the value of density of material will be

a) 0.04 b) 0.4

c) 40 d) 400

12. If the force is proportional to square of velocity, then the dimension of proportionality constant is [JEE-2000]

a) $[MLT^0]$ b) $[MLT^{-1}]$

c) $[ML^{-2}T]$ d) $[ML^{-1}T^0]$